Akash Prakash Sonth

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EDUCATION

Virginia Polytechnic Institute and State University (Virginia Tech) Blacksburg, VA, USA
M.S., Computer Engineering (concentration: Machine Learning), GPA: 3.88/4 *Aug 2021 - Jul 2023*

- Thesis: Enhancing road safety through machine learning for prediction of unsafe driving behaviors

Birla Institute of Technology and Science, Pilani (BITS Pilani) Pilani, India
B.E. (Hons.), Electrical and Electronics Engineering, GPA: 8.39/10 *Aug 2015 - Jul 2019*

- Thesis (at Brno University of Technology, Czechia): Vehicle density prediction and counting from monocular images
- Member, SAE Formula Student team: End-to-end design and integration of a Formula Student race car.

TECHNICAL SKILLS

Programming	Python, C#, MATLAB, SQL, LaTeX
Frameworks	PyTorch, PyG, Keras, TensorFlow, OpenCV, Open3D, scikit-learn
Robotics & Simulation	ROS, OpenStreetMap (OSM), Unity, Raspberry Pi
Systems & Cloud	Git, Docker, Linux, Bash, CI/CD, AWS (EC2, S3)
Agentic AI	Microsoft Agent Framework

EXPERIENCE

Emerson Electric (formerly **Aspen Technology**) Bedford, MA, USA
Data Scientist *Aug 2023 - Present*

- Developed an agentic framework for outlier detection, data quality assessment, and failure-relevant sensor selection in industrial time-series datasets, reducing analysis time from 2 hrs to under 5 mins per asset. *(Patent pending)*
- Designed an unsupervised equipment monitoring algorithm fusing expert-defined and generative-model-derived signatures, improving detection lead time by up to 10 hrs over production methods. *(Patent pending)*

Virginia Tech Transportation Institute Blacksburg, VA, USA
Graduate Research Assistant *Dec 2021 - Aug 2023*

- Designed a multi-stage perception pipeline to model traffic scenes as semantic scene graphs using Lanelet2 maps in OpenStreetMap format for lane-level topology.
- Developed TransformerConv and GINEConv based graph neural networks for intersection safety prediction, achieving 79% accuracy on low-resolution (320×240, 15 fps) videos.
- Led data acquisition for U.S. intersections, collecting synchronized drone and roadside video for SinD 2.0, a large-scale multimodal benchmark for traffic interaction and trajectory prediction. *(Public release forthcoming)*
- Adapted the Video Swin Transformer for explainable driver distraction monitoring by incorporating a visual dictionary of human actions, achieving 80.2% Top-1 accuracy on a large-scale low-quality dataset of 10k+ videos.
- Built a containerized Model Zoo for deployment of eye gaze estimation, multi-object tracking, and panoptic segmentation models.

Visual Computing Lab, Indian Institute of Science Bengaluru, India
Research Assistant *Feb 2020 - Apr 2021*

- Proposed an identity-free pose transformation method using non-parametric template cropping. Achieved state-of-the-art 93.6% Rank-1 accuracy on Market-1501 (32K+ images) using a two-stage self-supervised learning framework with a novel radial distance loss and ResNet-50 backbone.
- Implemented distributed multi-node, multi-GPU training using the NCCL backend, enabling 4x larger batch-sizes.

TEACHING EXPERIENCE

BITS Pilani | F312 Neural Networks and Fuzzy Logic - Teaching Assistant *Jan 2019 - May 2019*

PUBLICATIONS [Google Scholar]

- **Akash Sonth**, Lynn Abbott, Abhijit Sarkar. *"Intersection Safety Modeling Using Semantic Scene Graph and Graph Neural Network."* IEEE Intelligent Vehicles Symposium (IV), 2025. [Paper][Project Page]
- **Akash Sonth**, Abhijit Sarkar, Hirva Bhagat, Lynn Abbott. *"Explainable Driver Activity Recognition Using Video Transformer in Highly Automated Vehicle."* IEEE Intelligent Vehicles Symposium (IV), 2023. [Paper][Project Page]
- **Akash Sonth**, Harshavardhan Settibhaktini, Ankush Jahagirdar. *"Vehicle speed determination and license plate localization from monocular video streams."* International Conference on Computer Vision & Image Processing (CVIP), 2018. [Paper]
- Siddharth Seth, **Akash Sonth**, Anirban Chakraborty. *"Pose-Transformation and Radial Distance Clustering for Unsupervised Person Re-identification."* British Machine Vision Conference (BMVC), 2021. [Paper][Project Page]
- Hirva Bhagat, Sandesh Jain, Lynn Abbott, **Akash Sonth**, Abhijit Sarkar. *"Driver Gaze Fixation and Pattern Analysis in Safety Critical Events."* IEEE Intelligent Vehicles Symposium (IV), 2023. [PAPER]

AWARDS & FELLOWSHIPS

- Oral Presentation, IEEE Intelligent Vehicles Symposium (IV) (Top 7% of accepted papers), 2025
- IEEE Intelligent Transportation Systems Society (ITSS) flagship conference travel grant, 2025
- Google Research India Summer School (Computer Vision track), 2020 - one of 50 students selected nationwide.
- Virginia Tech, Center for Human-Computer Interaction (CHCI) conference award, 2023
- Virginia Tech, Graduate and Professional Student Senate (GPSS) conference award, 2023

SELECTED TALKS

- "Intersection Safety Modeling Using Semantic Scene Graph and Graph Neural Network." Invited talk at 2nd Workshop on Secure connected vehicles: Digital Twin, UAVs, and Smart Transportation, IEEE Intelligent Vehicles Symposium (IV), 2025.
- "Real-Time Risk Prediction at Signalized Intersection Using Graph Neural Network." Virginia Tech Transportation Institute (VTTI), Safe-D webinar, 2023 [Presentation]